

Options Lab: Client Solutions

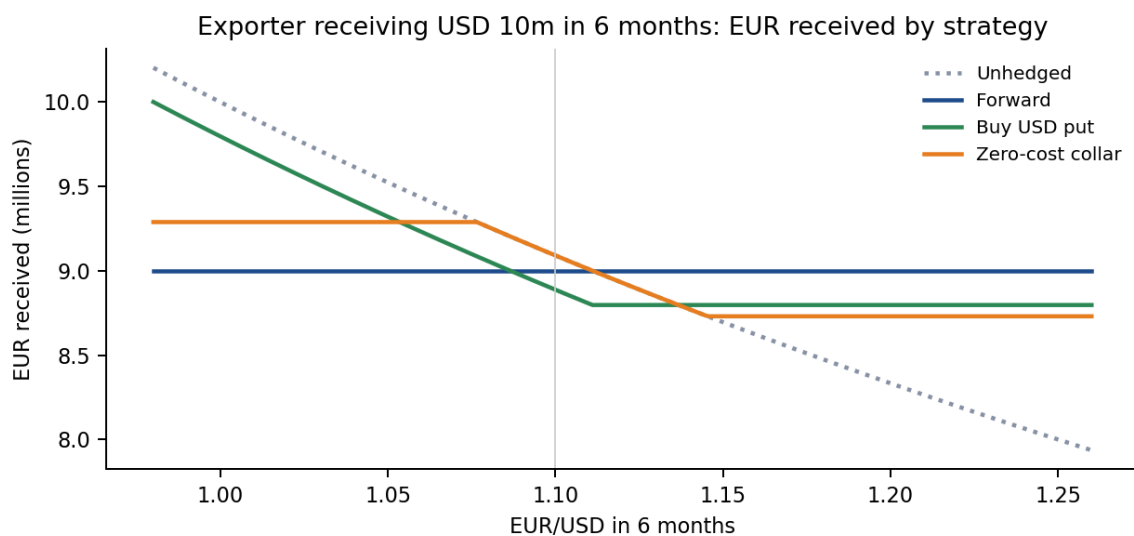
From pricing tools to client pitches · Eliote Demaret · EDHEC Business School

The pricing library (Black-Scholes / Garman-Kohlhagen, forwards, options) is used here to do what a sales desk does: understand a client's need, compare solutions, price them, and explain the trade-offs in plain language. **Market parameters are illustrative, not live quotes. These are not investment offers.**

Case 1: Corporate FX hedging for a eurozone exporter

Client need

A French industrial company will receive **USD 10m in 6 months** from a US customer. Its costs are in EUR, so it's exposed to a **fall in the dollar** (EUR/USD going up). Market: EUR/USD spot 1.1000, EUR rate 2%, USD rate 4%, 6-month vol 8%. The 6-month forward is **1.1111**: the higher-rate currency (USD) trades at a forward discount, which costs the exporter about 1% of carry.



Three solutions

Solution	How it works	Cost	Outcome
Forward	Sell USD 10m at 1.1111	No premium	Locks 9.00m EUR, no upside if USD rises
Buy a USD put (strike ATM-forward)	Right to sell USD at 1.1111	Premium 0.20m EUR (2.2% of notional)	Floor of 8.80m EUR after premium, keeps all the upside
Zero-cost collar	Buy USD put, sell USD call (strikes EUR/USD 1.1454 / 1.0765)	No premium	EUR received between 8.73m and 9.29m

EUR/USD at maturity	Unhedged	Forward	USD put	Zero-cost collar
1.00	10.00m	9.00m	9.80m	9.29m
1.05	9.52m	9.00m	9.32m	9.29m
1.10	9.09m	9.00m	8.89m	9.09m
1.15	8.70m	9.00m	8.80m	8.73m
1.20	8.33m	9.00m	8.80m	8.73m
1.25	8.00m	9.00m	8.80m	8.73m

Which solution for which client?

Client profile	Recommended
Budget rate must be guaranteed, no appetite for premium or complexity (most treasurers)	Forward
Wants protection but also a strong view that USD may rise, can pay premium	USD put
Wants protection with no upfront cost and accepts a capped upside	Zero-cost collar

Pitch in 3 sentences: "You have a natural long-dollar exposure, and a 10% fall in the dollar would cost you around EUR 0.8m. A forward locks your budget rate at 1.1111 for free. If you want to keep part of the upside, a zero-cost collar guarantees a worst case of 1.1454 while letting you benefit down to 1.0765, with no premium to pay."

Key points to explain: the forward rate isn't a forecast (it reflects the rate differential); "zero-cost" means no premium, not no cost (you give up upside); options require a credit line / hedge accounting documentation (IFRS 9).

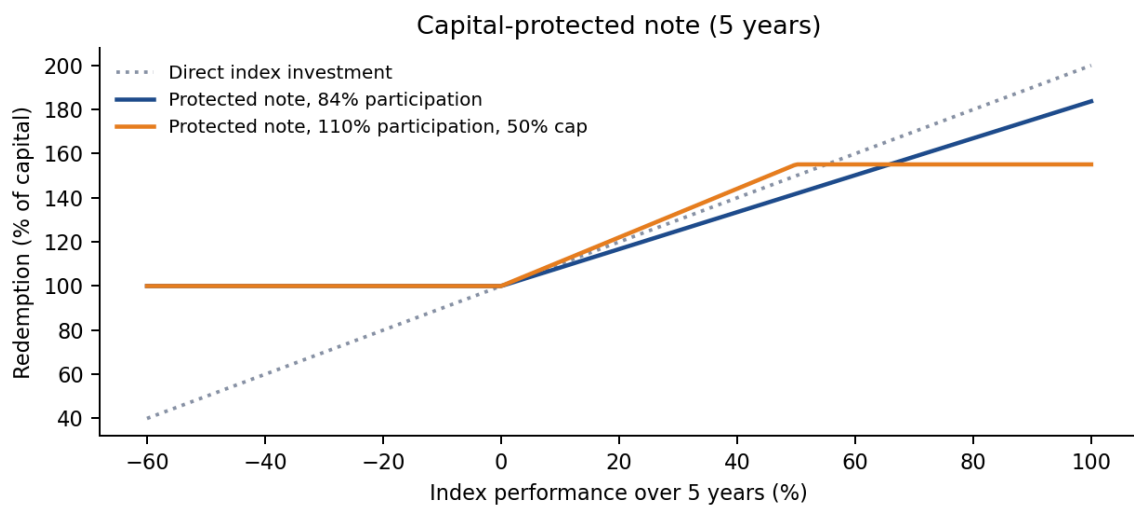
Case 2: Capital-protected note (private bank / retail investor)

Client need

A cautious investor wants exposure to the Euro Stoxx 50 for 5 years **without risking the capital**.
Assumptions: 5-year rate 2.5%, dividend yield 3%, vol 18%, distribution fees 1%.

How it is built

Building block	% of capital
Zero-coupon bond paying 100% in 5 years	88.2%
Fees / margin	1.0%
Budget left for options	10.8%
Price of a 5y ATM call	12.8%
Participation = budget / call price	84%
With a 50% cap (buy ATM call, sell 150% call)	110% participation, max gain 50%



Indicative term sheet

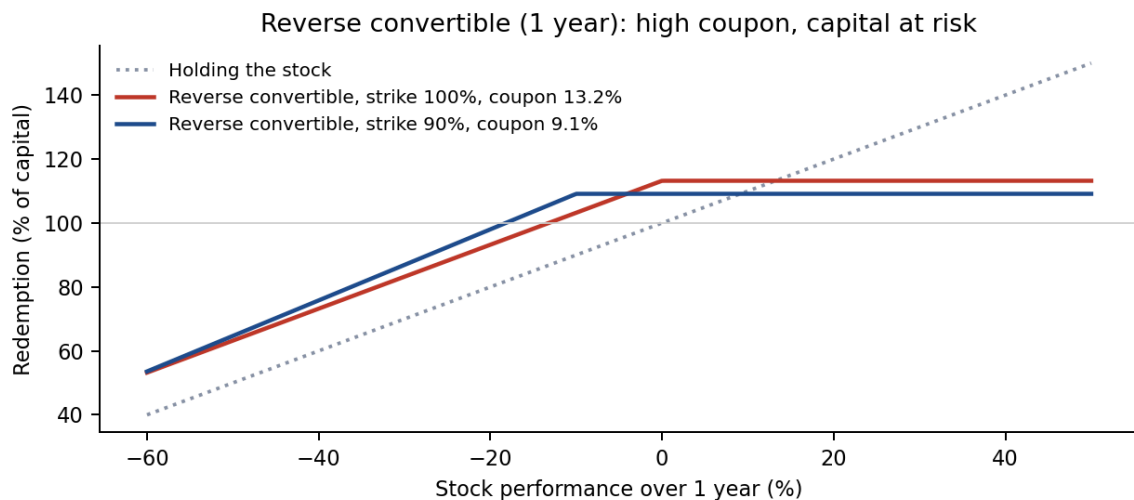
Underlying	Euro Stoxx 50 (price index)
Maturity	5 years
Capital protection	100% at maturity (subject to issuer credit risk)
Redemption at maturity	$100\% + 84\% \times \max(0, \text{index performance})$
Alternative with cap	$100\% + 110\% \times \min(\max(0, \text{perf}), 50\%)$

What drives participation: **higher rates** → cheaper zero-coupon → bigger option budget; **lower vol** → cheaper call; **higher dividends** → cheaper call (the investor doesn't receive them). Risks: issuer default, no protection before maturity, dividends lost.

Case 3: Reverse convertible (yield-seeking investor)

Client need

An investor is neutral to mildly bullish on a large-cap stock and wants a **high coupon** over 1 year. Assumptions: rate 2.5%, dividend yield 2%, stock vol 30%, fees 1%. The product is a **bond + a short put**: the investor is paid the put premium as an enhanced coupon.



Version	Put premium received	Coupon paid at maturity	Capital at risk if...
Strike 100%	11.4%	13.2%	stock ends below its initial level
Strike 90%	7.4%	9.1%	stock falls more than 10%

Indicative term sheet (strike 100%)

Coupon	13.2% paid at maturity, whatever happens
Redemption if final price $\geq$ strike	100% in cash
Redemption if final price $<$ strike	Delivery of shares ($\approx 100\% \times \text{final/strike}$)

Pitch: "If you're comfortable owning this stock 10% lower than today, the 90% version pays you a 9.1% coupon, against 2.5% on a deposit. The coupon is high because you're selling protection: if the stock falls more than 10%, you take the loss, partly cushioned by the coupon."

Why the coupon is high: it's the **volatility premium**. Higher vol $\rightarrow$ more expensive put $\rightarrow$ bigger coupon, but also a higher probability of losing capital. The investor is **short volatility**, like an option seller. The lower the strike, the smaller the coupon and the more protection.